HALIC UNIVERSITY / FACULTY OF ENGINEERING / DEPARTMENT OF COMPUTER ENGINEERING
2024 - 2025 ACADEMIC YEAR / SPRING SEMESTER
CENG – NUMERICAL ANALYSIS MIDTERM EXAM – 19/05/2025

Name & Surname :

Student Number :

Signature :

INSTRUCTIONS:

- The exam will start at 10:00 and will finish at 11:30. The total time of the exam is 90 minutes.
- This exam consists of four questions. Please explain your answers clearly.
- Use four digits after the decimal point.
- Answer the questions on the answer sheets provided to you. Any answers/solutions written on the question paper will not be taken into account.

Good luck!

QUESTIONS:

Q1. Find Lagrange interpolating polynomials that approximate $f(x) = \ln(1+x)$.

a. The linear interpolation polynomial $P_1(x)$ using the nodes $x_0 = 0$ and $x_1 = 1$. (15 pts)
b. The quadratic interpolation polynomial $P_2(x)$ using the nodes $x_0 = 0$, $x_1 = 1$ and $x_2 = 2$. (10 pts)
c. The cubic interpolation polynomial $P_3(x)$ using the nodes $x_0 = 0$, $x_1 = 1$, $x_2 = 2$ and $x_3 = 3$. (10 pts)

Q2. The following measurements were recorded in a study on the growth of trees:

Age (year):	7	10	15	20	25
Height (m):	5.2	7.8	9	10	10.6

The data is used for deriving an equation $H = H(Age)$ that can predict the height of the trees as a function of their age. Find the height of trees in the 16th year using the Newton's interpolating formula for the fourth order interpolation. Calculate the coefficients by using a divided difference table. (25 pts)

Q3. The following data is given for the stopping distance of a car on a wet road versus the speed at which it begins braking:

v (m/s):	12.5	25	37.5	50	62.5	75
d (ft):	20	59	118	197	296	420

a. Calculate the rate of change of the stopping distance at a speed of 62.5 mph using the backward difference formula. (10 pts)

b. Calculate an estimate for the stopping distance at 75 mph by using the results from part (a) for the slope and the central difference formula applied at the speed of 62.5 mph. (15 pts)



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Q4. The value of π can be calculated from the integral $\pi = \frac{1}{2}\int_{-1}^{1} \frac{4}{1+x^2} dx$. Approximate π using the composite trapezoidal method with six subintervals. (20 pts)

Q5. Evaluate the integral

$$\int_{0}^{2.4} \frac{2x}{1+x^2} dx$$

using the following methods.

a. Simpson's $\frac{1}{3}$ method. Divide the whole interval into six subintervals. (9 pts)
b. Simpson's $\frac{3}{8}$ method. Divide the whole interval into six subintervals. (9 pts)
c. The exact value of the integral is $I = \ln\left(\frac{169}{25}\right)$. Compare the results and discuss the reasons for the differences. (4 pts)

END OF THE QUESTIONS